

Yuanzhao Zhang

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APPOINTMENTS

<i>Department of Physics and Astronomy, University of Rochester</i> Simons Empire Assistant Professor	2026 –
<i>Santa Fe Institute</i> Omidyar Fellow · Independent position	2022 – 2026
<i>Center for Applied Mathematics, Cornell University & Santa Fe Institute</i> Schmidt Science Fellow · Advisor: Steven Strogatz	2020 – 2022
<i>IBM Thomas J. Watson Research Center</i> Research Intern · Advisor: Ruhong Zhou	2015

EDUCATION

<i>Northwestern University</i> Ph.D. Physics · Advisor: Adilson Motter	2020
<i>Northwestern University</i> M.Sc. Applied Mathematics	2015
<i>Zhejiang University</i> B.Sc. Mathematics, Chu Kochen Honors College	2014

RESEARCH INTERESTS

Nonlinear Dynamics, Complex Systems, Networks, Physics of AI

AWARDS AND FELLOWSHIPS

<i>Simons Empire Faculty Fellowship</i> Awarded by the Simons Foundation and Simons Foundation International to support early-career tenure-track faculty in mathematics and the basic sciences across New York State	2026
<i>APS Dissertation Award in Statistical and Nonlinear Physics</i> Awarded by American Physical Society	2021
<i>CSS Emerging Researcher Award</i> Awarded by Complex Systems Society	2021
<i>SIAM Student Paper Prize</i> Awarded by Society for Industrial and Applied Mathematics	2021
<i>Schmidt Science Fellowship</i> Postdoctoral fellowship supporting interdisciplinary scientists pivoting to a different field	2020
<i>Omidyar Fellowship</i> Independent postdoctoral fellowship awarded by the Santa Fe Institute	2020

GRANTS

<i>Closing the generalization gap of digital twins</i> DMS-2436231 · Funded by National Science Foundation · PI · \$ 269,979	2025 – 2028
<i>How do transcription factors control cell fate?</i> Schmidt Science Fellows Catalyst Grant · Funded by Schmidt Sciences · Co-PI · \$ 10,000	2022 – 2023
<i>Characterizing basins in high-dimensional landscapes</i> Lou Schuyler Research Grant · Funded by Santa Fe Institute · PI · \$ 15,000	2022 – 2023

PREPRINTS AND PUBLICATIONS

- I. León, R. Muolo, **Y. Zhang**, and M. Lucas, *Symmetry-based selection rules for higher-order interactions in coupled oscillators*, [arXiv:2606.04904](#)

- X. Cheng, W. Yuan, Z. Mu, **Y. Zhang**, Y. Yang, H. Wang, Z. Sun, and C. Liu, *Scaling world-model reinforcement learning through diffusion policy optimization*, [arXiv:2605.26282](#)
 - R. Delabays, **Y. Zhang**, F. Dörfler, and G. De Pasquale, *Data-driven control of hypergraphs: Leveraging THIS to damp noise in diffusive hypergraphs*, [arXiv:2511.08647](#)
 - **Y. Zhang**, K. M. Rock, and S. P. Cornelius, *Tempological control of network dynamics*, [arXiv:2510.10926](#)
 - P. S. Skardal, F. Battiston, M. Lucas, M. S. Mizuhara, G. Petri, and **Y. Zhang**, *Mixed higher-order coupling stabilizes new states*, [arXiv:2510.09387](#)
 - F. M. Brady*, **Y. Zhang***, and A. E. Motter, *Cluster synchronization in directed networks generate hierarchies*, [arXiv:2106.13220](#)
1. H. Hartle, P. L. Krapivsky, S. Redner, and **Y. Zhang**[†], *Anomalous scaling in redirection networks*, *Phys. Rev. E* **114** 034314 (2026)
 2. Z. G. Nicolaou, H. Cho, **Y. Zhang**, J. N. Kutz and S. L. Brunton, *Signature of glassy dynamics in dynamic mode decompositions*, *Phys. Rev. E* **113** L053301 (2026)
 3. B. H. Schlomann, W. S. DeWitt, **Y. Zhang**, and K. Shah, *Ignition criteria for trigger waves in cell signaling*, *PRX Life* **4** 023003 (2026)
 4. F. Battiston, C. Bick, M. Lucas, A. P. Millán, P. S. Skardal, and **Y. Zhang**[†], *Collective dynamics on higher-order networks*, *Nat. Rev. Phys.* **8** 146–159 (2026)
 5. **Y. Zhang**[†] and W. Gilpin, *Context parroting: A simple but tough-to-beat baseline for foundation models in scientific machine learning*, *ICLR 2026*
 6. X. Cheng, W. Yuan, Y. Yang, **Y. Zhang**, S. Cheng, Y. He, and Z. Sun, *Information shapes Koopman representation*, *ICLR 2026*
 7. P. Groisman, C. De Vita, J. F. Bonder, and **Y. Zhang**[†], *Size of the sync basin resolved*, *Phys. Rev. E* **112** L052201 (2025)
 8. D. A. Norton, **Y. Zhang**, and M. Girvan, *Learning beyond experience: Generalizing to unseen state space with reservoir computing*, *Chaos* **35** 103146 (2025)
 9. **Y. Zhang**[†], E. R. dos Santos, H. Zhang, and S. P. Cornelius, *How more data can hurt: Instability and regularization in next-generation reservoir computing*, *Chaos* **35** 073102 (2025)
 10. R. Delabays, G. De Pasquale, F. Dörfler, and **Y. Zhang**[†], *Hypergraph reconstruction from dynamics*, *Nat. Commun.* **16** 2691 (2025)
 11. **Y. Zhang** and W. Gilpin, *Zero-shot forecasting of chaotic systems*, *ICLR 2025*
 12. **Y. Zhang**[†], P. S. Skardal, F. Battiston, G. Petri, and M. Lucas, *Deeper but smaller: Higher-order interactions increase linear stability but shrink basins*, *Sci. Adv.* **10** eado8049 (2024)
 13. **Y. Zhang** and S. P. Cornelius, *Catch-22s of reservoir computing*, *Phys. Rev. Research* **5** 033213 (2023)
 14. Y. Huang, **Y. Zhang**, and R. Braun, *A minimal model of peripheral clocks reveals differential circadian re-entrainment in aging*, *Chaos* **33** 093104 (2023) · [Washington Post](#) · [BBC Science Focus](#)
 15. **Y. Zhang**^{*†}, M. Lucas*, and F. Battiston, *Higher-order interactions shape collective dynamics differently in hypergraphs and simplicial complexes*, *Nat. Commun.* **14** 1605 (2023)
 16. Z. Chen, T. Anglea, **Y. Zhang**, and Y. Wang, *Optimal synchronization in pulse-coupled oscillator networks using reinforcement learning*, *PNAS Nexus* **2** pgad102 (2023)
 17. **Y. Zhang** and S. H. Strogatz, *Basins with tentacles*, *Phys. Rev. Lett.* **127** 194101 (2021)
 18. **Y. Zhang**[†], V. Latora, and A. E. Motter, *Unified treatment of synchronization patterns in generalized networks with higher-order, multilayer, and temporal interactions*, *Commun. Phys.* **4** 195 (2021)
 19. **Y. Zhang**[†] and S. H. Strogatz, *Designing temporal networks that synchronize under resource constraints*, *Nat. Commun.* **12** 3273 (2021)
 20. **Y. Zhang**, J. L. Ocampo-Espindola, I. Z. Kiss, and A. E. Motter, *Random heterogeneity outperforms design in network synchronization*, *Proc. Natl. Acad. Sci. U.S.A.* **118** e2024299118 (2021)
 21. Y. Sugitani*, **Y. Zhang***, and A. E. Motter, *Synchronizing chaos with imperfections*, *Phys. Rev. Lett.* **126** 164101 (2021)
 22. **Y. Zhang** and A. E. Motter, *Mechanism for strong chimeras*, *Phys. Rev. Lett.* **126** 094101 (2021)
 23. M. Feng, Y. Song, S. H. Chen, **Y. Zhang**, and R. Zhou, *Molecular mechanism of secreted Amyloid- β precursor protein in binding and modulating GABA_BR1a*, *Chem. Sci.* **12** 6107-6116 (2021)

24. **Y. Zhang**[†] and A. E. Motter, *Symmetry-independent stability analysis of synchronization patterns*, *SIAM Rev.* **62** 817–836 (2020) · [Winner of the SIAM Student Paper Prize](#)
25. **Y. Zhang**, Z. G. Nicolaou, J. D. Hart, R. Roy, and A. E. Motter, *Critical switching in globally attractive chimeras*, *Phys. Rev. X* **10** 011044 (2020)
26. B. Li, **Y. Zhang**, X. Meng, and R. Zhou, *Zipper-like unfolding of dsDNA caused by graphene wrinkles*, *J. Phys. Chem. C* **124** 3332–3340 (2020)
27. J. D. Hart*, **Y. Zhang***, R. Roy, and A. E. Motter, *Topological control of synchronization patterns: Trading symmetry for stability*, *Phys. Rev. Lett.* **122** 058301 (2019) · [Quanta Magazine](#)
28. **Y. Zhang** and A. E. Motter, *Identical synchronization of nonidentical oscillators: When only birds of different feathers flock together*, *Nonlinearity* **31** R1–R23 (2018)
29. **Y. Zhang**, T. Nishikawa, and A. E. Motter, *Asymmetry-induced synchronization in oscillator networks*, *Phys. Rev. E* **95** 062215 (2017)
30. G. Duan*, **Y. Zhang***, B. Luan, J. K. Weber, R. W. Zhou, Z. Yang, L. Zhao, J. Xu, J. Luo and R. Zhou, *Graphene-induced pore formation on cell membranes*, *Sci. Rep.* **7** 42767 (2017)
31. **Y. Zhang**, J. K. Weber, and R. Zhou, *Folding and stabilization of native-sequence-reversed proteins*, *Sci. Rep.* **6** 25138 (2016)
32. Z. Lin and **Y. Zhang**, *Stirring by multiple cylinders in potential flow*, *J. Fluid Mech.* **794** 552 (2016)
33. Z. Gu*, **Y. Zhang***, B. Luan, and R. Zhou, *DNA translocation through single-layer boron nitride nanopores*, *Soft Matter* **12** 817 (2016)
34. **Y. Zhang***, C. A. Jimenez-Cruz*, J. Wang, B. Zhou, Z. Yang, and R. Zhou, *Bio-mimicking of proline-rich motif applied to carbon nanotube reveals unexpected subtleties underlying nanoparticle functionalization*, *Sci. Rep.* **4** 7229 (2014)
35. Y. Tu, H. Lu, **Y. Zhang**, T. Huynh, and R. Zhou, *Capability of charge signal conversion and transmission by water chains confined inside Y-shaped carbon nanotubes*, *J. Chem. Phys.* **138** 015104 (2013)

* equal contributions

†corresponding author

INVITED TALKS

Schmidt Interdisciplinary Science Summit	Durham 2026
Mechanical Engineering Seminar @ Johns Hopkins University	Baltimore 2026
Physics Seminar @ UC Berkeley	Berkeley 2026
Computational & Applied Mathematics Colloquium @ University of Chicago	Chicago 2026
Center for Computational Biology @ Flatiron Institute	New York 2025
Physics Colloquium @ National University of Singapore	Singapore 2025
SPMS Seminar @ Nanyang Technological University	Singapore 2025
Dynamics Days Europe	Thessaloniki 2025
BeyondTheEdge Seminar	Online 2025
Seminar @ University of Applied Sciences and Arts of Western Switzerland	Sion 2025
Seminar @ Complexity Science Hub	Vienna 2025
SIAM Conference on Applications of Dynamical Systems	Denver 2025
Center for Interdisciplinary Studies Seminar @ Westlake University	Hangzhou 2025
WTI Symposium on Cognition and Computation @ Yale University	New Haven 2025
PSU-Purdue-UMD Joint Seminar on Mathematical Data Science	Online 2025
EEB Seminar @ Princeton University	Princeton 2025
Oden & Neuroscience Seminar @ UT Austin	Austin 2025
Data Science & Math Seminar @ UNC Chapel Hill	Chapel Hill 2025
KIAS-KU workshop on Theoretical Challenges in Network Science	Seoul 2024
TopoNets @ NetSci	Quebec City 2024
Applied Dynamics Seminar @ University of Maryland	College Park 2024
Workshop on Modelling and Mining Complex Networks as Hypergraphs	Toronto 2024
Machine Learning and Dynamical Systems Seminar @ Alan Turing Institute	Online 2024
Physics Seminar @ University of Maryland	College Park 2024

<i>Complex Systems Seminar @ University of Michigan</i>	Ann Arbor 2024
<i>Mathematics Seminar @ UCSD</i>	San Diego 2023
<i>CNLD Seminar @ UT Austin</i>	Austin 2023
<i>NetPLACE Seminar</i>	Online 2023
<i>Collective Behavior Gordon Research Conference</i>	Newry 2023
<i>CNLS Seminar @ Los Alamos National Laboratory</i>	Los Alamos 2023
<i>Complexity Conclave</i>	Santa Fe 2023
<i>SIAM Conference on Applications of Dynamical Systems</i>	Portland 2023
<i>Mathematics Seminar @ UIUC</i>	Champaign 2022
<i>Statistical Physics Seminar @ Chinese Academy of Sciences</i>	Online 2022
<i>Prize Talk @ APS March Meeting</i>	Chicago 2022
<i>Physics Seminar @ UC Berkeley</i>	Berkeley 2022
<i>Applied Interdisciplinary Math Seminar @ University of Michigan</i>	Online 2022
<i>Complex Systems & Statistical Mechanics Seminar @ ICTP-SAIFR</i>	Online 2021
<i>Systems Neuroscience and Complexity Seminar @ University of Sydney</i>	Online 2021
<i>Prize Talk @ SIAM Annual Meeting</i>	Online 2021
<i>Oberseminar Dynamics @ Technische Universität München</i>	Online 2021
<i>Applied & Computational Mathematics Seminar @ Dartmouth</i>	Online 2020

TEACHING

<i>Statistical Mechanics</i>	University of Rochester · 2027
<i>Higher-Order Networks</i>	NetSci School · 2022
<i>Dynamical Systems</i>	Complex Systems Summer School · 2022

SERVICE

<i>Panelist, National Science Foundation</i>	2025
<i>Co-organizer, SFI Working Group “Dynamical systems and graph theory approaches for digital twins”</i>	2025
<i>Ad hoc reviewer, Army Research Office</i>	2023
<i>Council member, Complex Systems Society</i>	2021 – 2024
<i>Co-organizer, Chicago Area SIAM Student Conferences</i>	2017 – 2020
<i>Co-organizer, Bridging the Gap Seminar</i>	2017 – 2020
<i>President, Northwestern University Student Chapter of SIAM</i>	2018 – 2020
<i>Vice President, Northwestern University Student Chapter of SIAM</i>	2017 – 2018

MENTORING

<i>Declan Norton</i>	2026 –
Postdoc at University of Rochester	
<i>Emmie Fitz-Gibbon</i>	2026 –
Undergraduate student at Brown University	
<i>Katherine Li</i>	2022 – 2023
Undergraduate student at Minerva University · Current position: graduate student in physics at Johns Hopkins University	
<i>Fiona M. Brady</i>	2018 – 2020
Undergraduate student at Northwestern University · Current position: graduate student in mathematics at Princeton University	